

Synthetic CO2 Room Transmitter - Technical Datasheet

Supplier model CO2-TX-2K - Revision D

Product summary

Synthetic wall-mount nondispersive-infrared CO2 transmitter for building ventilation applications. This datasheet is provided as a hardware evidence source; the baseline simulation uses an ideal concentration sensor but retains the same measurement range.

Parameter	Value
Measurement principle	NDIR, self-calibrating
CO2 range	0 to 2000 ppm
Accuracy at 25 C	$\pm(40 \text{ ppm} + 3\% \text{ of reading})$
Resolution	1 ppm
Typical response time (T63)	90 s
Analog output	0-10 V = 0-2000 ppm
Digital output	BACnet MS/TP, CO2 object in ppm
Supply	24 VAC/VDC
Operating temperature	0 to 50 C
Recommended mounting	1.1 to 1.7 m above finished floor
Fault indication	Value flagged invalid on internal diagnostics

Project integration note

The physical device reports ppm. ICD-IAQ-03 defines the internal simulation/controller concentration signal as trace-substance mass fraction in kg/kg, followed by a normalization gain. Therefore a SysML model may legitimately represent both a sensor display quantity in ppm and an internal feedback quantity in kg/kg, provided the conversion/interface is explicit.